

PRODUCT REQUIREMENT DOCUMENT (PRD)

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Project Name: AI-Powered Vendor and Procurement Agent

Version: 1.0

1. Project Overview

- **Problem Statement:** In modern corporate procurement, teams are overwhelmed by unstructured data. Companies often request quotes from at least three different vendors to ensure competitive pricing, but these quotes arrive as static, multi-page PDFs with varying layouts. Manual comparison is slow, labor-intensive and highly susceptible to human oversight. Crucially, traditional methods focus solely on price, often missing “Red Flags” such as poor vendor reputation, financial stability or legal news, until after a contract is signed. This lack of a 360-degree view leads to value leakage, risky partnerships and inefficient purchasing cycles.
- **Target Domain:** Enterprise Procurement and Supply Chain Management, specifically focusing on the “Source-to-Pay” workflow. This system is designed for organizations that manage high volumes of vendor quotes and require a faster, data-backend method to compare costs, risks and market benchmarks simultaneously.
- **Proposed Solution Summary:** The AI-Powered Vendor and Procurement Agent is an autonomous intelligence layer that transforms procurement from a manual data-entry task into a high-level curation process. By integrating pdfplumber for PDF text extraction and [Z.AI](#) GLM for intelligent data parsing, alongside a live Adversarial Research engine, the agent reads PDFs like a human, verifies vendor credibility via web search and correlates these findings against historical market benchmarks stored in JSONBin.io. The solution provides a unified Comparison Dashboard where the AI acts as a “Senior Auditor”, highlighting the best value-to-risk ratio. Finally, it automates the administrative closing of the deal by generating professional Purchase Orders and confirmation emails, allowing the user to shift their role from data processor to strategic decision-maker.

2. Background & Business Objective

- **Background of the Problem:** For decades, the procurement process has followed a rigid, manual and linear pathway. A procurement officer typically receives dozens of vendor quotes via email, manually downloads the PDFs and transcribes the data into a central spreadsheet. They then conduct independent, manual Google searches to verify vendor credibility and cross-reference prices with historical records or market trends. This process is heavily reliant on copy-paste labor, expensive ERP software licenses and the specific expertise of senior buyers. Because this process is so tedious, many companies default to the lowest price without performing deep due diligence on hidden risks or long-term vendor stability.
- **Importance of Solving This Issue:** By automating the transition from raw PDF to comparative insight, the organization can accelerate the procurement cycle from days to minutes. Solving this issue is critical because it eliminates the “Manual Error Gap”, where typos in unit prices lead to significant financial loss. Furthermore, by breaking the information barrier through AI-driven Adversarial Research, the system uncovers red flags that would otherwise require hours of investigative work. This enables the company to pivot from a reactive “price-taking” stance to a proactive “risk-managed” strategy, ensuring that every dollar spent is optimized for both value and security.
- **Strategic Fit / Impact:** The project aligns with the modern AI-Native Enterprise movement, where generative AI is used not just as a chatbot, but as an operational backbone. By integrating [Z.AI](#)’s GLM and DuckDuckGo web scraping, the system creates an intelligent bridge between unstructured external data (PDFs/Web News) and internal financial systems ([JSONBin.io](#) history). This integration transforms the procurement department into a data-driven hub, significantly reducing operational overhead while providing a high-fidelity audit trail for corporate compliance and financial transparency.

3. Product Purpose

3.1. Main Goal of the System: To provide a “Cognitive Procurement Partner” that automates the transition from raw vendor data to strategic purchasing decisions. The system aims to eliminate manual data entry, uncover hidden vendor risks through automated research and ensure that every Purchase Order is backed by a 360-degree analysis of cost, market benchmarks and reliability.

3.2. Intended Users (Target Audience):

- Procurement Officers/Buyers
- Finance Managers/Controllers
- Compliance and Risk Officers
- Supply Chain Directors
- Operations Managers
- IT/System Administrators

4. System Functionalities

4.1. Description: The system operates as an Autonomous Procurement Intelligence Engine that processes unstructured vendor data through natural language and document vision. Using a Document-to-Decision model, the agent captures complex pricing structures, identifies external risks and synthesizes multi-vendor comparisons to transform raw quotes into actionable procurement contracts.

4.2. Key Functionalities:

- **Zero-Touch Ingestion:** Automatically extracts line items, tax structures and terms from various PDF layouts, converting static text into structured database entries.

- **Adversarial Risk Research:** Proactively searches the live web for vendor news, lawsuits and financial sentiment to populate a “Red Flag” dashboard.
- **Multi-Factor Synthesis:** Correlates quote data with market benchmarks and historical internal performance to provide a weighted “Best Value” recommendation.
- **Generative Fulfillment:** Drafts professional, legally-compliant Purchase Order (POs) and vendor acceptance emails tailored to the final agreed-upon terms.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: The core engine utilizes [Z.AI](#) GLM. It was selected for its strong reasoning capabilities and reliable structured JSON output. Combined with pdfplumber for text extraction, the GLM parses unstructured quote text into structured data fields. A robust JSON extraction parser handles code fences, truncated responses and nested objects to ensure reliable output for financial data extraction. It has the ability to perform spatial document reading (understanding table hierarchies in PDFs) and native JSON-mode output makes it ideal for high-precision financial data extraction where zero margin for error is permitted.

4.3.2. Prompting Strategy: We employ a Multi-Step Agentic Prompting strategy. This is because a single prompt is insufficient for complex procurement; Step 1 focuses on Extraction, Step 2 on Risk Cross-Referencing and Step 3 on Synthesis. Each step uses a dedicated system prompt sent as a separate GLM chat completion call, ensuring that the AI evaluates risk factors independently of price, preventing the model from ignoring red flags just because a price is low.

4.3.3. Context & Input Handling: The system handles large inputs by truncating PDF text to 4000 characters before sending to the GLM. If extraction fails at this length, a shorter 1500-character fallback is attempted. This ensures the system stays within token limits while extracting the most critical pricing data. Text is trimmed from the beginning, and the GLM prompt instructs the model to extract what it can from the available portion.

4.3.4. Fallback & Failure Behavior: To prevent "financial hallucinations", the system utilizes a confidence scoring mechanism. Each extracted item receives a confidence score (0.0 to 1.0) from the GLM. If extraction fails on the first attempt, the system triggers one automatic retry with shorter input text. If extraction still fails, the system returns a placeholder item with confidence 0.0 and vendor name "EXTRACTION_FAILED", flagging it for the user. The user reviews all extracted data in Step 2 of the dashboard with confidence bars visible, ensuring human-in-the-loop verification before proceeding.

5. User Stories & Use Cases

- **User Stories:**

- As a procurement officer, I want to upload multiple vendor PDFs at once so that the system can automatically extract and compare pricing without me having to type data into a spreadsheet.
- As a finance manager, I want to see a "Market Benchmark" comparison for every line item so that I can identify if we are overpaying relative to current industry rates.
- As a risk and compliance lead, I want the system to scan for external news and lawsuits regarding a vendor so that I can flag potential "Red Flags" before we commit to a contract.
- As an operations manager, I want a drafted Purchase Order and vendor email generated automatically after I make a selection so that I can close the deal instantly without administrative delays.

- **Use Cases (Main Interactions):**

Quote-to-Insight Flow: User uploads three competing vendor PDF quotes, then the ingestion service ([Z.AI](#)) parses the unstructured documents, identifies unit prices and terms, and populates the Comparison Dashboard. As a result the user sees a side-by-side grid highlighting the lowest cost, fastest delivery and best value options.

Adversarial Risk Discovery: User clicks “Analyze Risk” for a specific vendor, then the research agent executes web searches via DuckDuckGo HTML scraping, analyzing search result snippets for that vendor name. The GLM then identifies red flags and positive notes from the search results. If no web results are found, the GLM provides a knowledge-based assessment. As a result, the system renders a "Risk Profile" with red flags and positive notes.

Decision Synthesis and PO Generation: User selects their preferred vendor and clicks “Finalize Selection”, then the fulfillment service pulls the specific quote data and pre-approved legal templates. As a result, the system generates a finalized PDF Purchase Order and a drafted confirmation email ready for one-click dispatch.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction).*



6. Features Included (Scope Definition)

- A file upload workspace where users can upload multiple unstructured PDF quotes simultaneously for automated text extraction via pdfplumber and data normalization via GLM.

- Adversarial Risk Dashboard where a dedicated research panel automatically displays "Red Flags" and positive notes for every vendor, pulled from DuckDuckGo web search results and analyzed by the GLM.
- A dynamic, grid-based UI that highlights discrepancies between vendors, such as price anomalies, delivery lead times and risk scores.
- A feature that compares extracted line-item prices against internal historical data and external market averages to detect overspending.
- An automated document fulfillment tool that converts a selected quote into a finalized, branded Purchase Order (PDF) and a ready-to-send vendor email.
- An interactive interface that displays AI-extracted values with confidence scores, allowing users to review and verify data before proceeding to research and analysis.
- A persistent quote history stored in JSONBin.io that records every extracted vendor quote with timestamps, confidence scores and raw text hashes, providing transparency for future reference.

7. Features Not Included (Scope Control)

- Limited to digital-text PDFs; handwritten or extremely low-resolution physical scans are out of scope.
- Core analysis and "Red Flag" research are optimized for English-language documents and news sources.
- This version focuses on standalone CSV/PDF exports; direct bidirectional API sync with SAP/Oracle is reserved for Phase 2.
- The system provides data for negotiation but does not currently act as an autonomous chatbot to haggle with vendors directly.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The estimated cost is RM0.20 to RM0.60 per procurement session (assuming 3 vendor quotes of 5 pages each). Thus, each processed quote generates a raw_text_hash stored in JSONBin.io for future deduplication. The MVP currently re-processes uploads on each session. Additionally, web research returns up to 8

search results per query to prevent context window bloat, with results staggered at 0.5-second intervals to avoid rate limiting.

- **Technical Constraints:** For export formats, initial PO generation is limited to standard PDF and JSON exports. Direct integration with legacy ERP systems (SAP/Oracle) via API is not supported in the MVP; users must use manual CSV uploads for data bringing. For source format, only digital PDFs are accepted. The system cannot reliably parse low-resolution mobile photos or handwritten quotes due to the risk of financial data misinterpretation.
- **Performance Constraints:** High-fidelity reasoning for the Brain requires a multi-step agentic chain, which can take 15-30 seconds per session. Users are limited to comparing a maximum of five vendors simultaneously to prevent session timeouts. Other than that, to manage [Z.AI](#) API throughput, there is no hard upload cap; throughput is limited by GLM API rate limits and [JSONBin.io](#) write rate (1 write/sec).
- **User Input:** Since the system is an assistant, not an approver, it cannot finalize a Purchase Order without explicit human signature and approval. Moreover, the quality of the Adversarial Research depends on the user providing the correct Legal Entity Name of the vendor.

9. Risks & Questions Throughout Development

- **Financial Accuracy (The “Hallucination” Risk):** How do we ensure the AI does not hallucinate a price that looks realistic but does not match the source PDF? We address this via confidence scoring (0.0 to 1.0) and human-in-the-loop review in Step 2.
- **Real-time Data Freshness:** What happens if the “Red Flag” research engine misses a very recent news event (e.g., a lawsuit filed this morning)?
- **Contextual Ambiguity:** What will happen if the AI is unable to contextualize a specific industry term or a non-standard discount (e.g., “Early Bird Rebate of 2% if paid in 10 days”)?

- **Data Privacy:** Are we inadvertently training the public model on our company's sensitive procurement data?